

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

Expected May 2028

College Park, MD

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Writing Java logic that decides how much EC2 capacity Amazon Redshift reserves for operations like patching, setting upper and lower bounds that keep availability high without overspending, and backtesting the policies

Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Rebuilt the legacy tool developers use to create test-data workflows as a React and TypeScript Electron app with a visual React Flow editor and Jest-tested validation, and **90%** of 30+ surveyed developers preferred it
- Because the 270K+ row PostgreSQL workflow table had strict access limits, precomputed popular workflows into a category tree that the Claude skill searches first, with Redis caching full workflows to cut database queries
- Cut AI workflow generation time **65%** to under 15s by splitting one large Claude skill into smaller sub-skills that call Python tools for search and validation, after CloudWatch and OpenSearch latency logs showed where time went

UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Deployed a space-topic research tool for Amazon Leo's public-policy team and set it up for handoff with AWS CDK infrastructure-as-code and GitHub Actions CI/CD, so Leo engineers new to the project now maintain it on AWS Fargate
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda
- Automated retraining for a production Amazon Bedrock model that sorts research documents by importance: on an EventBridge schedule a larger LLM grades it, and it is fine-tuned only when those grades shift past a threshold

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- For a PhD-led robotics project targeting ICLR, built a Python pipeline that draws AnyGrasp candidates from multi-camera images as Open3D overlays and tuned prompts so a vision-language model picks the right grasp more often
- Cut research-environment setup time **30%** with uv dependency pinning and Docker, and added psutil time and memory profiling and Pytest unit tests to help the team find bottlenecks

TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Co-founded a 110+ member student product studio that has shipped 5 live products, and personally built Redis caching in front of the Prisma ORM queries for one, Tortuga, a class scheduler with **20,000+ visits** in its first 30 days
- Building a RAG meal-suggestion feature for the official UMD app (GPT-4o mini, ChromaDB + SentenceBERT search); **93%** of generated meals rated satisfactory by internal testers

Projects

IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine and replayed **50.9M** real IEX exchange messages through it without ever resyncing to the exchange's book, matching 99,122 of 99,123 fills and 100% of top-of-book prices
- Compared B+ tree and LSM skip-list storage for the engine's event logs on identical 2M-record workloads: the B+ tree read **12.8x** faster and answered range queries 22x faster, at the cost of 27% more disk
- Sustained **1.26M events/sec** end-to-end over the full replay, with price levels in a std::map of FIFO queues and a hash map from order ID to queue position for O(1) cancels

Technical Skills

Languages: Python, Java, C++, C, TypeScript, JavaScript, SQL, Bash

Cloud & DevOps: AWS (Lambda, Kinesis, Fargate, EC2, API Gateway, EventBridge, CloudWatch), AWS CDK (infrastructure as code), Docker, GitHub Actions (CI/CD)

Backend & Full-Stack: REST APIs, Node.js, React, Electron, PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, ChromaDB

AI/ML: LLMs (Claude, Bedrock, GPT-4o mini), retrieval-augmented generation (RAG), fine-tuning and evaluation, PyTorch

Practices & Tools: data structures and algorithms, unit testing (Pytest, Jest), profiling (perf, psutil), Linux, Git, Jira, Agile

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)